

Numerical Analysis Project: Project Report Guidelines

Report Overview

In addition to submitting your functional Python code, you are required to submit a formal project report. In numerical analysis, communicating *why* an algorithm behaves the way it does is just as important as writing the code. Your report should be written in the style of a standard scientific paper, utilizing clear mathematical typesetting and professional formatting.

Required Document Structure

Your report must be 4 to 7 pages in length and include the following sections in this exact order:

1. Title, Name, and Abstract

- **Title & Name:** Provide a descriptive title for your project and include your full name.
- **Abstract:** A concise summary (100–150 words) of the entire project. Briefly state the problem being solved, the mathematical methods used (Standard Newton vs. Gauss-Newton), and the primary conclusion regarding convergence and real-world noise.

2. Introduction & Mathematical Background

Clearly explain the problem without simply copying the assignment prompt.

- **The Problem:** Explain how GPS determines a position using trilateration, pseudorange, and clock bias.
- **The Math:** Define the system of nonlinear equations $F(\mathbf{u})$.
- **The Algorithms:** Explain the mathematical difference between finding a root using the standard Multivariate Newton Method ($J\Delta\mathbf{u} = -F$) and minimizing the sum of squares using the Gauss-Newton Method ($(J^T J)\Delta\mathbf{u} = -J^T F$).
- **Requirement:** You must explicitly write out the formulas for the i -th row of the Jacobian matrix $J(\mathbf{u})$ using proper mathematical formatting.

3. Implementation & Setup

Discuss how you structured the problem computationally.

- **Initialization:** What initial guess $\mathbf{u}_0$ did you use and why?

- **Stopping Criteria:** Define the absolute or relative error tolerances you used to terminate your ‘while’ loop (e.g., $\|\Delta \mathbf{u}_k\|_2 < \epsilon$) and state your maximum iteration limit.
- **Linear Algebra:** Briefly mention which computational tools (e.g., `numpy.linalg.lstsq`) were used to solve the linear systems safely.

4. Results & Visualizations

This section proves that your implementation works.

- **Final Coordinates:** Provide a clear table displaying your final computed (x, y, z) position and clock bias d for **both** the 4-satellite (Newton) and N -satellite (Gauss-Newton) scenarios.
- **Convergence Data:** Include a table tracking the step size (error proxy) at each iteration k , alongside your calculated Experimental Order of Convergence (EOC) p .
- **Log-Error Plot:** You **must** include a plot showing the iteration number k on the x-axis and $\log_{10}(\|\Delta \mathbf{u}_k\|_2)$ on the y-axis. Both methods should be plotted to compare their convergence slopes. All figures must have labeled axes, a legend, and a caption.

5. Discussion & Analysis

Demonstrate your critical thinking and understanding of numerical methods by answering the following:

- **The “Perfect” Illusion:** When using exactly 4 satellites, your algorithm likely found a point where the residual was exactly zero ($F(\mathbf{u}) = \mathbf{0}$). Explain why this mathematically “perfect” root resulted in a geographically inaccurate location compared to the N -satellite method.
- **Convergence Behavior:** Did both methods achieve the quadratic convergence rate ($p \approx 2$)? Discuss how the noisy atmospheric data affected the convergence rate of the Gauss-Newton method during its final iterations.
- **Algorithmic Limitations:** What would happen to your Jacobian matrix if two of the chosen satellites were in the exact same position in the sky? (Hint: Consider matrix conditioning and singularity).

6. References

- You must include **at least 4 trustworthy references**.
- Acceptable sources include numerical analysis textbooks, peer-reviewed academic journals, and official technical documentation (e.g., official Python/NumPy documentation or GPS mathematical specifications).
- Avoid citing generic websites, wikis, or blog posts. Use a consistent citation format (such as IEEE, APA, or MLA) throughout the text and in the reference list.